

Model Answer Sheet of SPPU Insem

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[6009]-327

T.E. (Computer/A.I&D.S) (Insem.)
CLOUD COMPUTING (Elective - II)
(2019 Pattern) (Semester - II) (310254C)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks
- 4) Assume suitable data if necessary.

- Q1)** a) Define Cloud Computing & Explain the importance of Cloud Computing? [5]
b) Explain Cloud System Architecture suitable diagram? [5]
c) Discuss Cloud Deployment Models with suitable diagram? [5]

OR

- Q2)** a) Describe the Trends in Cloud Computing? [5]
b) Draw and Explain Cloud Service Model? [5]
c) Explain Seven-step model of migration into a Cloud? [5]

- Q3)** a) Explain how the Cloud Data Management Works? [5]
b) How the HDFS Architecture works? Explain it with suitable diagram? [5]
c) Explain the Storage Area Network with suitable diagram? [5]

OR

- Q4)** a) Explain the features of GFS Architecture? [5]
b) Describe Data Intensive Technologies for Cloud Computing? [5]
c) Identify the advantage and disadvantageous of Direct Attached Storage? [5]



Unit I: Introduction to Cloud Computing

Q.1] Define Cloud Computing & Explain the importance of Cloud Computing?



Study material provided by: Vishwajeet Londhe

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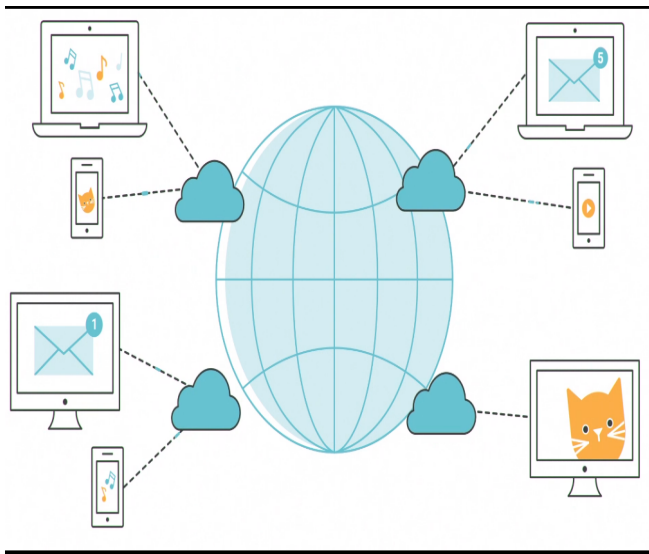
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Answer:

Cloud computing means storing and accessing data and programs over the internet instead of your computer's hard drive

Cloud Computing is a network of remote servers hosted on the internet for storing and retrieving data.



Example :from reference book

● 21st Feb 2011, Sometime in ICC Cricket World Cup, India Furious cricket fans slammed organizers of the World Cup on Monday as the official ticketing website crashed amid a scramble for 1,000 tickets available for the final.(Source: Times of India [Magazine])

The manager of ICC posted a message in his Facebook fan page:

We are facing absolutely unprecedented amounts of traffic from all over the world with hundreds of millions of people hitting at once. Some of you may have trouble accessing the site. It seems that cricket fever has surpassed all anticipations and expectations. Please bear with us as our global network team works on bringing you the tickets you all have been waiting for. (Source: Kyazoonga FB page)

The majority of us will answer as follows: **Add additional servers to balance the load.**

- **Before cloud computing**, companies had to store all their data and software on **their own hard drives and servers**.
- The **bigger the company, the more storage they needed**.
- This way of treating data is **not scalable** at speed.
- **Cloud technology** means that companies can **scale and adapt at speed and scale, accelerate innovation, streamline operations**, and **reduce costs**.

1. Scalability :Scalability refers to the expansion of infrastructure to handle an increased load for every application.

2. Flexibility:Cloud computing promotes flexibility in the workplace. It allows the employees to be flexible enough by accessing the data from home or on a holiday.

3. Saving Costs:the time to spend on computing, storage, and networking is reduced.

4. Disaster Recovery:Disaster Recovery or DRaaS is a cloud computing service model that helps an organization back up its data over the cloud with the help of an IT infrastructure third-party cloud computing environment.

5.Easy Accessibility:Cloud computing is an easy-to-access computing system.

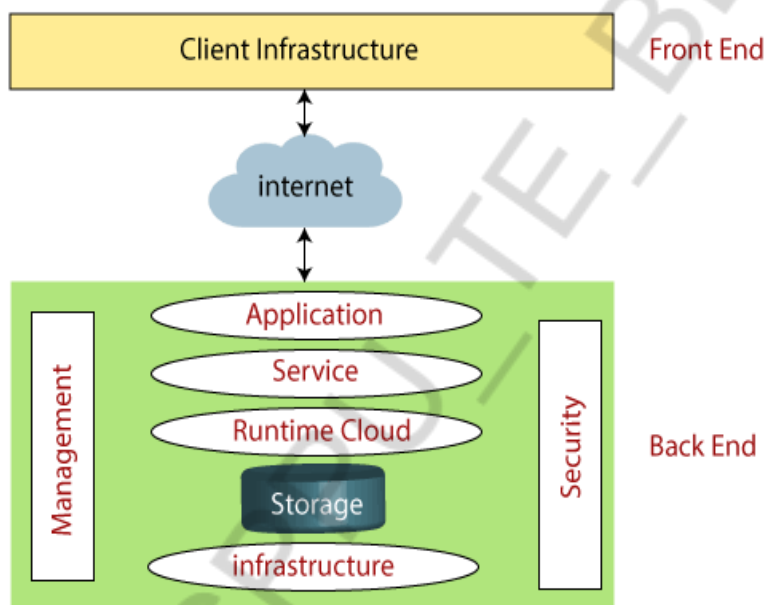
6.Higher Level of Security:Cloud computing provides a higher level of security when it comes to both public and private clouds.

7.Automatic Updates:One of the biggest reasons why cloud computing is interesting is because it provides automatic updates

Q.1.b) Explain Cloud System Architecture suitable diagram?

Answer:Cloud architectures can be viewed as a collection of different functionalities and capabilities.

Cloud architecture basically describes the **communication** between the client and server via network, which is mostly the internet.



Components of Cloud Computing Architecture

There are the following components of cloud computing architecture -

1. Client Infrastructure

Client Infrastructure is a Front end component. It provides **GUI (Graphical User Interface)** to interact with the cloud.

2. Application

The application may be any software or platform that a client wants to access.

3. Service

A Cloud Services manages that which type of service you access **according to the client's requirement.**

Cloud computing offers the following **three type of services:SaaS,IaaS,PaaS**

4. Runtime Cloud

Runtime Cloud provides the **execution and runtime environment** to the virtual machines.

5. Storage

Storage is one of the most important components of cloud computing. It provides a huge amount of storage capacity in the cloud to store and manage data.

6. Infrastructure

It **provides services** on the **host level, application level, and network level.** Cloud infrastructure includes hardware and software components such as servers, storage, network devices, virtualization software, and other storage resources that are needed to support the cloud computing model.

7. Management

Management is used to **manage components** such as **application, service, runtime cloud, storage, infrastructure, and other security issues** in the backend and establish coordination between them.

8. Security

Security is an **in-built back end component** of cloud computing. It implements a security mechanism in the back end.

9. Internet

The Internet is medium through which front end and back end can interact and communicate with each other.

Q.1 c] Discuss Cloud Deployment Models with suitable diagrams?

Answer: Deployment model depending on how much data you want to store and who has access to the Infrastructure

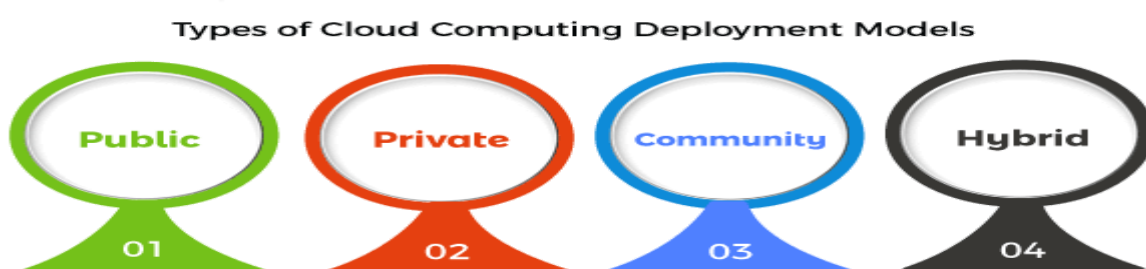




Figure 6.6 Public Cloud

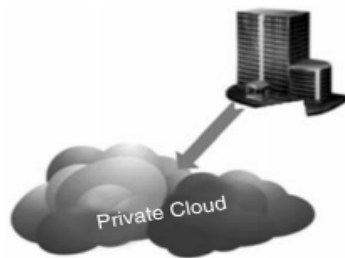


Figure 6.5 Private Cloud

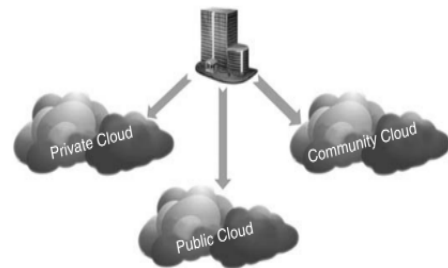
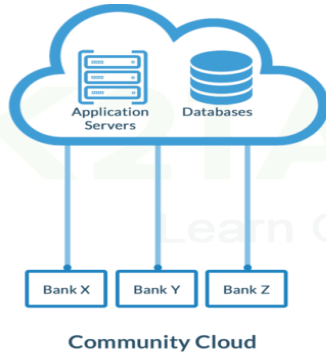


Figure 6.7 Hybrid Cloud



- **Private cloud:** A **private cloud** is a server, data center, or distributed network **wholly dedicated to one organization**.
- **Public cloud:** A **public cloud** is a service run by an external vendor that may include servers **in one or multiple data centers**. Unlike a private cloud, **public clouds are shared by multiple organizations**. Using virtual machines, individual servers may be shared by different companies, a situation that is called "**multitenancy**" because multiple tenants are renting server space within the same server.
- **Hybrid cloud:** hybrid cloud deployments combine public and private clouds, and may even include on-premises legacy servers. An organization may use their private cloud for some services and their public cloud for others, or they may use the public cloud as backup for their private cloud.
- **Community Cloud:** It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business.
- The infrastructure of the community could be shared between the organization which has shared concerns or tasks.
- It is generally managed by a third party or by the combination of one or more organizations in the community.

Q.2.a) Describe recent Trends in Computing?

Answer: The following are some of the biggest cloud computing trends

1. Edge Computing

2. AI and ML
3. Multi and Hybrid Cloud Solution
4. Cloud Gaming
5. Cloud Security and Resilience
6. Serverless Computing
7. Kubernetes
8. Blockchain
9. IOT

1. Edge Computing:

In edge computing, data is stored, processed, and analyzed geographically closer to its source

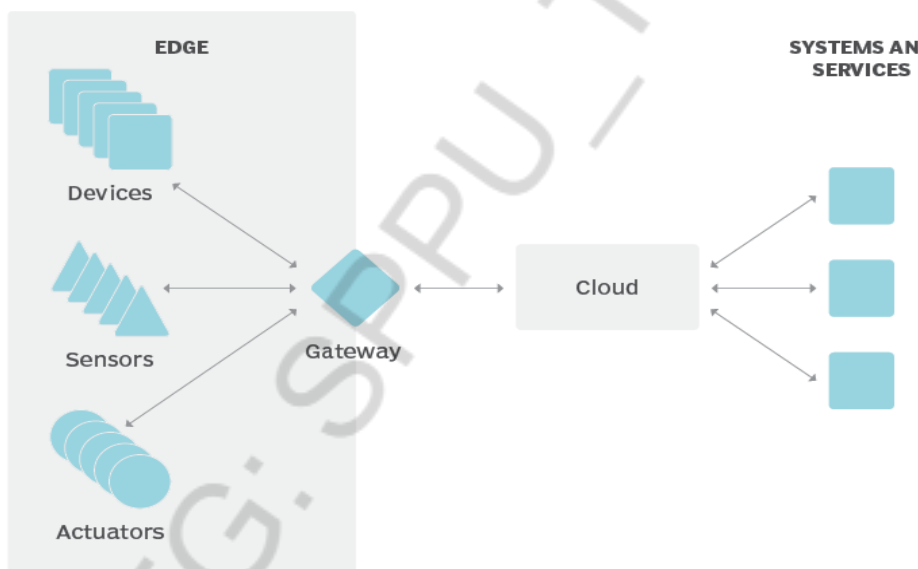
Edge computing applications that process and analyze data at the network's edge are expanding as a result of the increasing use of 5G.

Edge computing also allows companies to streamline operations, automate procedures, boost performance, and, most importantly, accelerate their digital transformation.

Most cloud platforms will feature a sizable number of distributed cloud services that operate at the edge,

Cloud computing is the act of running workloads within clouds, while edge computing is the act of running workloads on edge devices.

An edge device is any piece of hardware that controls data flow at the boundary between two networks.



2. AI and ML

Artificial Intelligence and Machine Learning are two technologies that are closely related to cloud computing.

They are a solution for managing massive volumes of data to improve tech company productivity.

The key trends that are likely to emerge in this area include **increased automation and self-learning capabilities**, greater data security and privacy, and more personalized cloud experiences.

3. Multi and Hybrid Cloud Solution

A lot of enterprises have adopted **multi-cloud and Hybrid IT strategy which combines on-premise**, dedicated private clouds, several public clouds, and legacy platforms.

The reason is that businesses now understand that data management involves combining a variety of appropriate cloud deployment strategies **rather than relying solely on one cloud platform or architecture**.

They offer a combination of **public and private clouds dedicated to a specific company** whose data is key business driver, such as insurance, banks, etc.

multi and hybrid cloud solutions will be among the **top cloud computing trends** in the coming years.

4. Cloud Gaming

Video gaming services are provided by Microsoft, Sony, Nvidia, and Amazon.

But streaming video games require higher bandwidth and can be possible only with high-speed internet access.

Cloud gaming will become a significant industry in 2023 with the introduction of 5G.

5. Cloud Security and Resilience(Adapt well to change)

Organizations and companies face a new range of **cyber security threats** **when they migrate to the cloud**.

Cloud migration offers many benefits, efficiencies, and conveniences but **presents several security risks**.

investing in cyber security and **building resilience against everything from data loss** on global business will become a priority during the coming year.

“**security-as-a-service**” providers and **AI and predictive technology** to detect risks before they cause issues.

6. Serverless Computing:

Serverless computing allows developers **to run code without worrying about infrastructure management**.

This trend is gaining popularity because it reduces the cost and complexity of running applications in the cloud.

7. BlockChain:

Blockchain is a linked list of blocks containing records and keeps growing as users add to it.

The blockchain provides a **robust and permanent audit trail of every transaction** in the proper order

It can process vast amounts of data and exercise control over documents economically and securely.

8. IOT

It is a technology that maintains connections between computers, servers, and networks.

IoT functions as a **mediator**, ensuring successful communication and **assisting in data collection from remote devices for efficient operations.**

9. Kubernetes

Kubernetes, also known as K8s, is an open-source system for automating deployment, scaling, and management of containerized applications.

Kubernetes are rapidly evolving and will continue to be major players in cloud computing trends over the next few years.

Q2.b) Draw and Explain Cloud service models.

Answer:

There are the following three types of cloud service models -

1. **Software** as a Service (SaaS)
2. **Platform** as a Service (PaaS)
3. **Infrastructure** as a Service (IaaS)

1. Software as a Service (SaaS)

Provider of SaaS has full administrative rights for its application and responsible for activities such as deployment, maintenance and update.

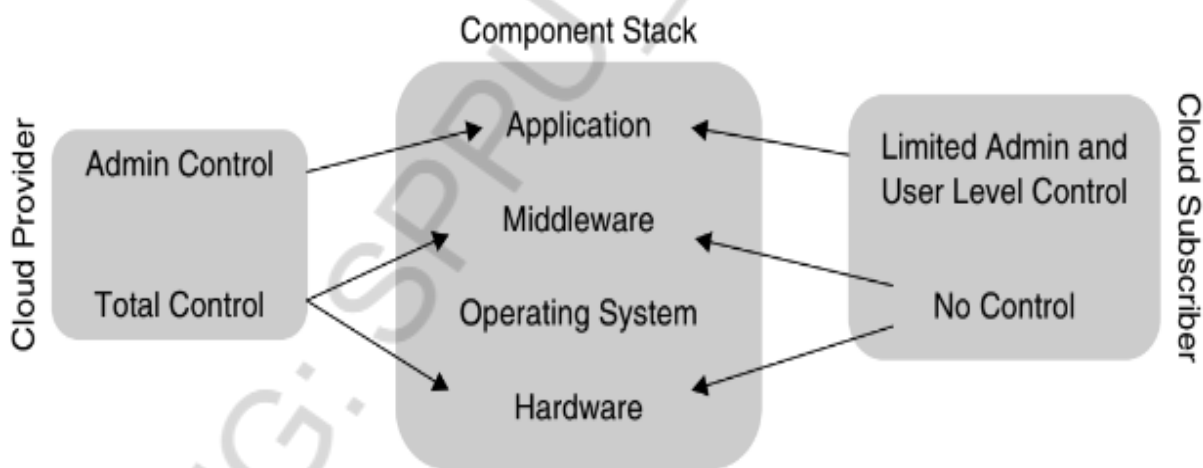


Figure 4.3 SaaS Component Stack and Scope of Control

- Figure 4.3 shows the levels of rights between the subscriber and the provider,
- From the figure, it is clear that a **cloud provider has total control over the hardware, middleware and operating system.**
- It also has **administrative control over the application residing in the server.**
- **Cloud subscriber subscribes the service,** it has limited admin and user level control.
- **Cloud users do not have control over the OS or the hardware.**

- **SaaS subscribers can be individual users**, users from organizations and users from enterprises.
- If the focus is on improving of the business, SaaS is the best option.

There are the following characteristics of SaaS -

- Managed from a central location
- Hosted on a remote server
- Accessible over the internet
- Users are not responsible for hardware and software updates. Updates are applied automatically.
- The services are purchased on the pay-as-per-use basis

Example: BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, and GoToMeeting.

2. Platform as a Service (PaaS)

- PaaS is service, where application/software can be build, tested and deployed as a single unit.
- PaaS is useful for application builders, developers, deployers and testers.

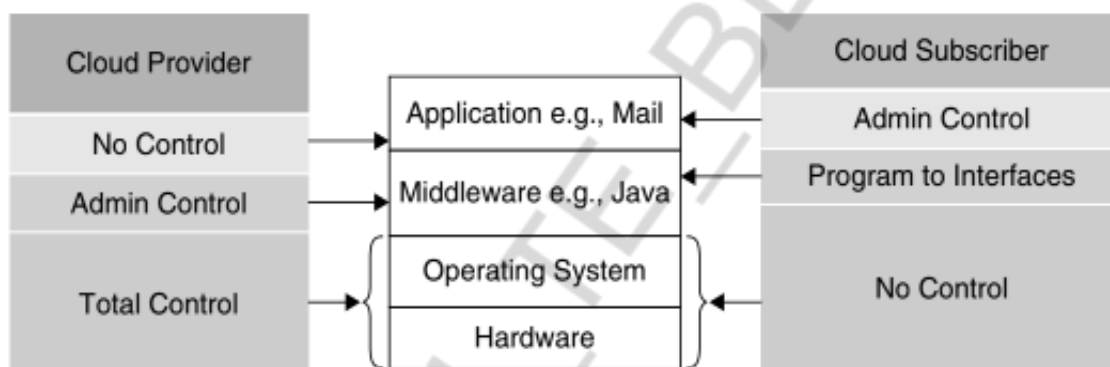


Figure 4.4 PaaS Component Stack and Scope of Control

- From the figure, we can understand that the **cloud provider has total control over the hardware and operating system, admin control over the middleware and no control over the application.**
- A **cloud subscriber** subscribes to the services and has full admin rights over the application deployed and minimal rights over the middleware.
- **Cloud users** do not have control over the OS or the hardware.

3. Infrastructure as a Service (IaaS)

- When the customer requires an end-to-end infrastructure such as computer resources, storages and network, he/she can opt for IaaS.
- The usage fee is billed at CPU hour, size (GB) of data accessed or stored/hour, bandwidth consumed, etc.

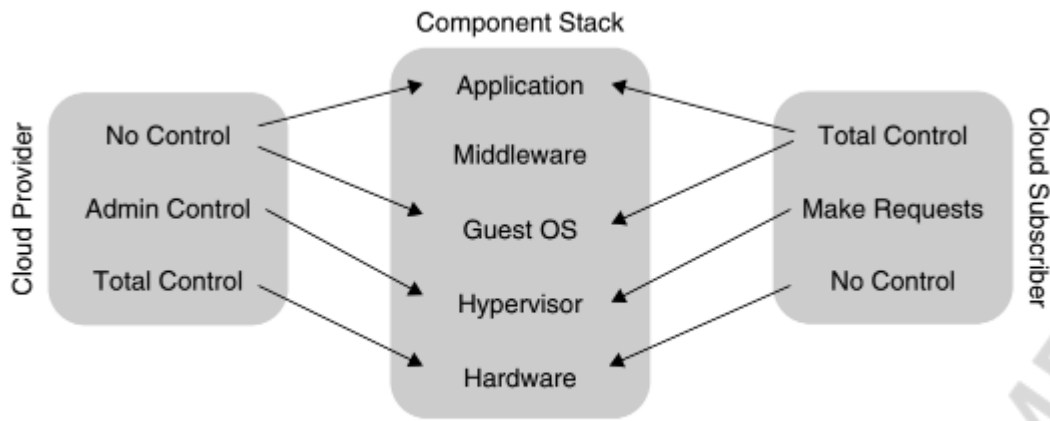


Figure 4.5 IaaS Component Stack and Scope of Control

-
- From the figure, it is clear that cloud provider has total control only over the hardware and has admin rights for virtualization part, that is, hypervisor.
- He/she has no control over the application, middleware and guest operating system. Cloud subscriber subscribes the service and has full admin rights for the application deployed, middleware and the OS. Cloud users can make requests to hypervisor but don't have control over the hardware.
- Enterprises comprising of many servers can act as an IaaS provider such as Facebook, Orkut and Twitter.

Q.2 C] Explain Seven-step model of migration into a Cloud?

Answer:

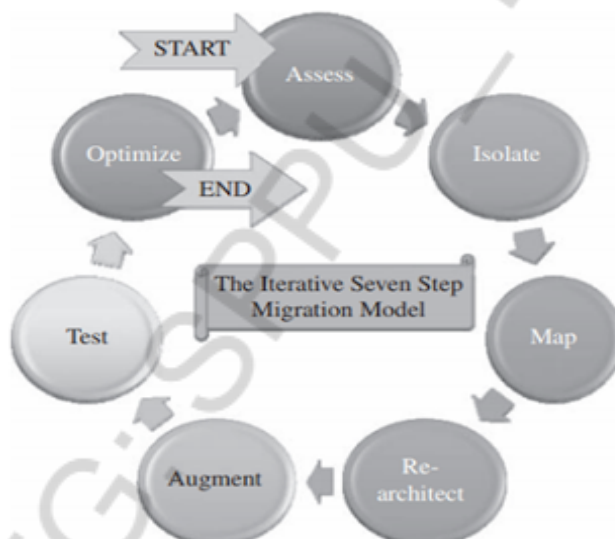


FIGURE 2.5. The iterative Seven-step Model of Migration into the Cloud. (Source: Infosys Research.)

1.ASSESSMENT

- Migration starts with an assessment of the **issues relating to migration, at the application, code, design, and architecture levels.**
- Moreover, **assessments are also required** for tools being used, functionality, test cases, and configuration of the application.

- The proof of concepts for migration and the corresponding pricing details will help to assess these issues properly

2. ISOLATE

- The second step is the **isolation of all the environmental and systemic dependencies of the enterprise application within the captive data center.**
- These include **library, application, and architectural dependencies.**
- This step results in a better understanding of the complexity of the migration.

3. MAP

- A mapping construct is generated **to separate the components** that **should reside in the captive data center** from the ones that will go into the cloud.

4. RE-ARCHITECT

- It is likely that a substantial part of the application has to be re-architected and implemented in the cloud.
- **This can affect the functionalities of the application and some of these might be lost.**
- It is possible to approximate **lost functionality using cloud runtime support API.**

5. AUGMENT

- The features of cloud computing service are used to augment the application

6. TEST

- Once the augmentation is done, the application needs to be validated and tested.
- This is to be done using a test suite for the applications on the cloud.
- New test cases due to augmentation and proof-of-concepts are also tested at this stage.

7. OPTIMISE

- The test results from the last step can be mixed and so **require iteration and optimization.**
- It may take several optimizing iterations for the migration to be successful.
- **It is best to iterate through this seven step model as this will ensure the migration to be robust and comprehensive.**

Q.3.a] Explain how the Cloud data management works?

Answer:

- For Effective data management in cloud storage requires several key considerations,

including: Security, Data backup and recovery, Data privacy, Scalability and flexibility, Cost-effectiveness etc

- Cloud storage should incorporate new services according to change of time.
- For cloud storage, a standard document is placed by SNIA, Storage Industry Resource Domain Model (SIRDm).
- Figure shows the SIRDm model which uses CDMI standards.
- SIRDm model adopts [three metadata](#):
- [storage metadata, data metadata and user metadata](#).
- By using these metadata, cloud storage interface **can offer services** without adding unnecessary complexity in managing the data.

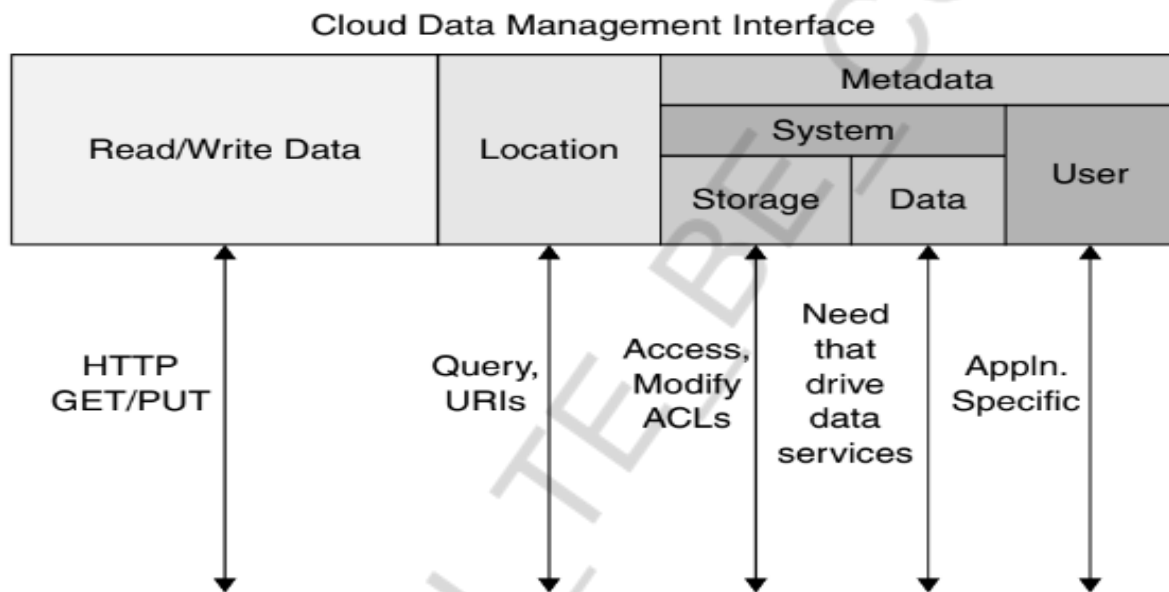


Figure 12.1 Cloud Storage Usage of SIRDm Model

- [User metadata](#) is used by the cloud to find the data objects and containers.
- [Storage system metadata](#) is used by the cloud to offer basic storage functions like assigning, modifying and access control.
- [Data system metadata](#) is used by the cloud to offer data as a service based on user requirements and controls the operation based on that data.
- **The functions in CDMI are:**
- To create, retrieve, update and delete objects in a cloud, the cloud data management interface (CDMI) is used.
-
- Cloud storage offerings are discovered by clients
- Management of containers and the data
- Sync metadata with containers and objects
- CDMI is also used to manage containers, domains, security access and billing information.

- CDMI standard is also **used as protocols** for accessing storage.
- CDMI defines how to manage data and also ways of storing and retrieving it.
- **‘Data path’** means how data is stored and retrieved.
- **‘Control path’** means how data is managed.
- CDMI standard supports both data path and control path interface.

Q3.b] How the HDFS Architecture Works? Explain it with Suitable Diagram?

Answer:

- HDFS Hadoop distributed file system.
- HDFS is an Open source component of the Apache Software Foundation that manages data.
- HDFS has scalability, availability, and replication as key features. Name nodes, secondary name nodes, data nodes, checkpoint nodes, backup nodes, and blocks all make up the architecture of HDFS
- **Hadoop File System** :A distributed file system designed to run on commodity hardware is known as Hadoop Distributed File System (HDFS).
- In HDFS, files are stored in **blocks ranging from 64 MB to 1024 MB**.
- The default size is 64 MB.
- The blocks will be distributed across the cluster and replicated for fault tolerance.

NameNode

All the blocks on DataNodes are handled by NameNode, which is known as the master node. It performs the following functions:

1. Monitor and control all the DataNodes instances.
2. Permits the user to access a file.
3. Stores all of the block records on a DataNode instance.

NameNode

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1. Monitor and control all the DataNodes instances.
2. Permits the user to access a file.
3. Stores all of the block records on a DataNode instance.

4. EditLogs are committed to disk after every write operation to Name Node's data storage. The data is then replicated to all the other data nodes, including Data Node and Backup Data Node. In the event of a system failure, EditLogs can be manually recovered by Data Node.
5. All of the DataNodes' blocks must be alive in order for all of the blocks to be removed from the data nodes.

There are two kinds of files in NameNode: FsImage files and EditLogs files:

Secondary NameNode

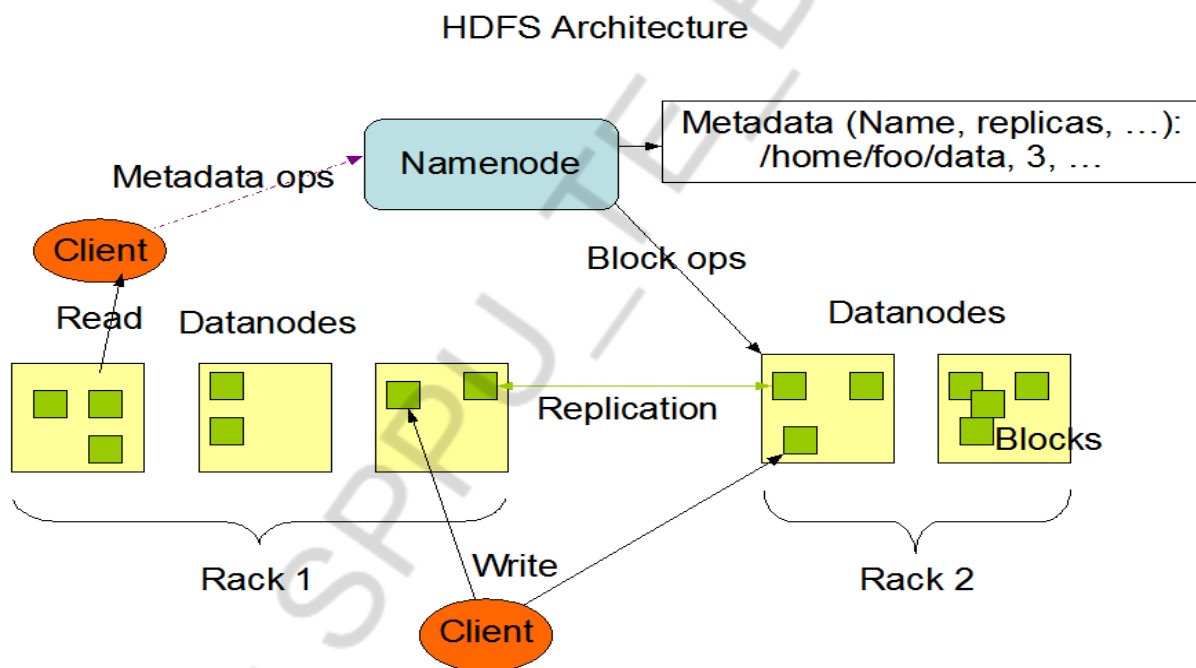
When NameNode runs out of disk space, a secondary NameNode is activated to perform a checkpoint. The secondary NameNode performs the following duties.

1. It stores all the transaction log data (from all the source databases) into one location so that when you want to replay it, it is at one single location. Once the data is stored, it is replicated across all the servers, either directly or via a distributed file system.

DataNode

Every slave machine that contains data organises a DataNode. DataNode stores data in ext3 or ext4 file format on DataNodes. DataNodes do the following:

1. DataNodes store every data.
2. It handles all of the requested operations on files, such as reading file content and creating new data, as described above.
3. All the instructions are followed, including scrubbing data on DataNodes, establishing partnerships, and so on.



Q3.c) Explain the storage area Network with suitable diagrams?

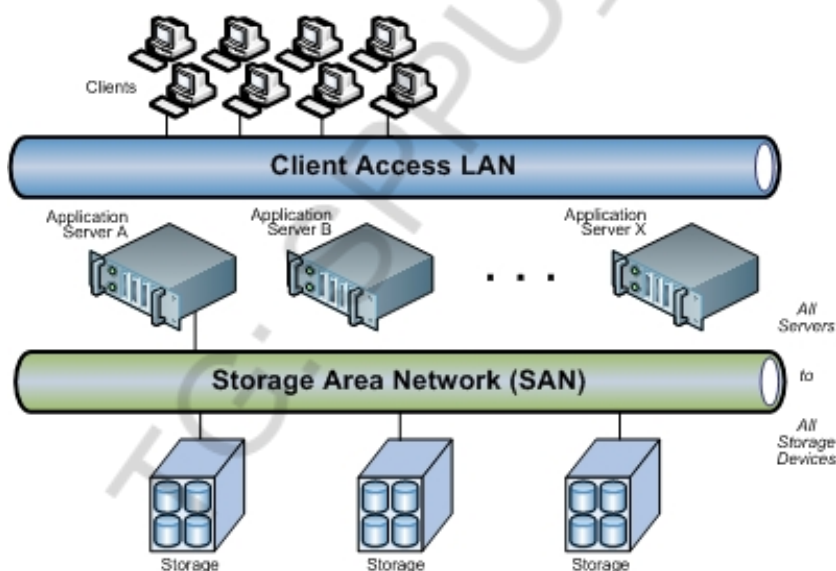
- A **Storage Area Network** (SAN) is a specialized, high-speed network that provides network access to storage devices.
- SANs are typically composed of hosts, switches, storage elements, and storage devices that are interconnected using a variety of technologies, topologies, and protocols.
- Traditionally, only a limited number of storage devices could attach to a server, limiting a

network's storage capacity.

- But a **SAN introduces** networking flexibility **enabling one server, or many heterogeneous servers** across multiple data centers, **to share a common storage utility**.
- **Advantage of SAN**
- Simplified storage administration
- Disk mirroring
- Low cost of storage management
- Instant and real-time information
- Ability to boot itself and expand the storage capacity
- SAN is not directly attached to any particular server or network, SAN can be shared by all

Disadvantage of SAN

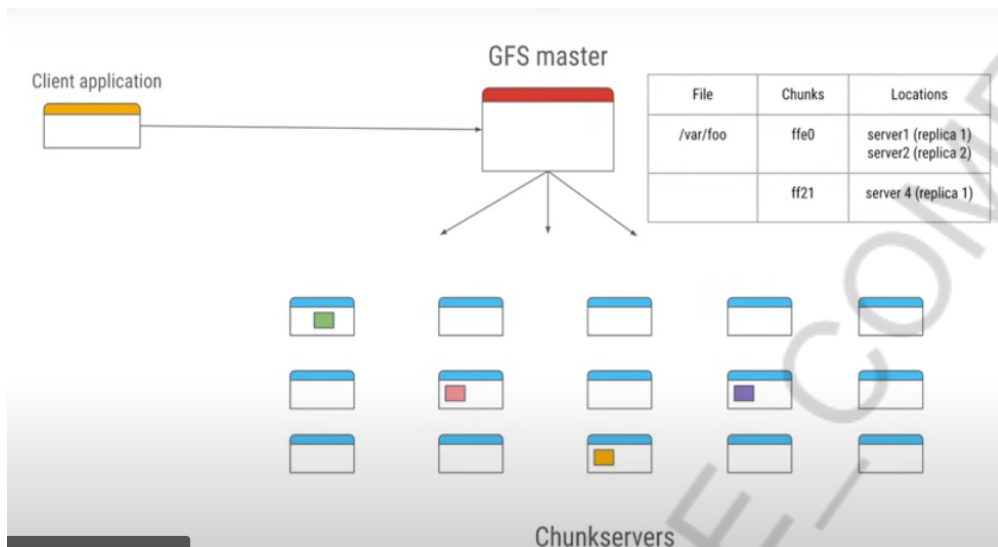
- If client computers need intensive data transfer then SAN is not the right choice. SAN is good for low data traffic
- More expensive
- It is very hard to maintain
- As all client computers share the same set of storage devices so **sensitive data can be leaked**. It is preferable not to store confidential information on this network.
- Poor implementation results in a performance bottleneck
- Not affordable for small business
- Require a high-level technical person



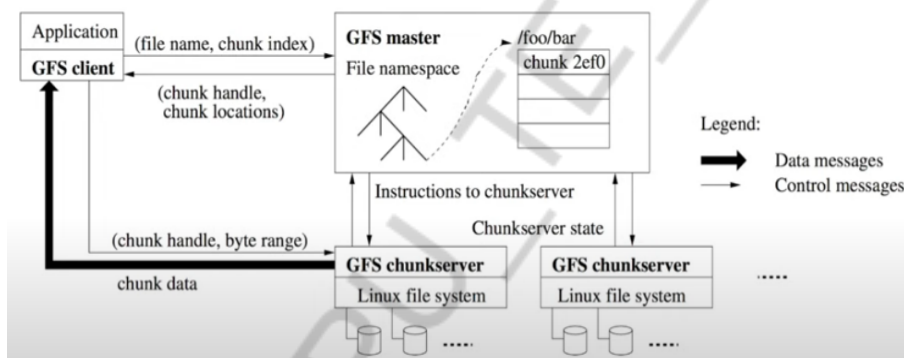
Q.4] a) Explain features of GFS Architecture?

Answer:Google file System

- It is **scalable distributed file system** for large distributed data-intensive applications.
- Developed by google
- Single file is not stored into single server,files are divided into multiple chunk
- GFS master only read the metadata of file

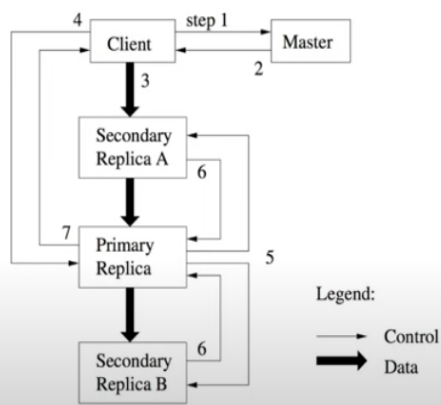


Reads



Writes

1. Ask for locations to write
2. Get replicate locations
3. Write data to closest replica.
4. Request commit to primary
5. Primary instructs order of writes to secondaries
6. Secondaries acknowledge
7. Primary ack to client



File system typically provide mechanism for **reading,writing ,modifying,deleting** or organising files in **folders and directories**

Gluster File System:

Users no longer locked with legacy storage platforms which are costly and monolithic.

GlusterFS gives users the ability to deploy scale-out, virtualized storage, centrally managed pool of storage.

Attributes of GlusterFS include scalability and performance, high availability, global namespace, elastic hash algorithm, elastic volume manager, gluster console manager, and standards-based.

Q.4.b] Describe data-intensive Technologies for Cloud Computing?

Answer:

Data-intensive computing is a related type of computing which use parallelism concept for processing large volumes of data, called big data.

Parallel processing approaches are divided into two types: compute-intensive and data intensive.

Processing Approach

Data-intensive computing platforms use a parallel computing approach.

This approach combines multiple processors and disks as computing clusters connected via high-speed network.

The data that are needed to be processed are independently done by computing resources available in the clusters.

There are many common characteristics of data-intensive

The principle mechanism used for collection of the data and programs or algorithms to perform the computation

Programming model used

Reliability and availability

Scalability of both hardware and software

For data-intensive computing an array of system architectures have been implemented.

Architecture for data-intensive computing 1. MapReduce 2. HPCC

1. MapReduce

MapReduce concept which is developed by Google and available as open-source

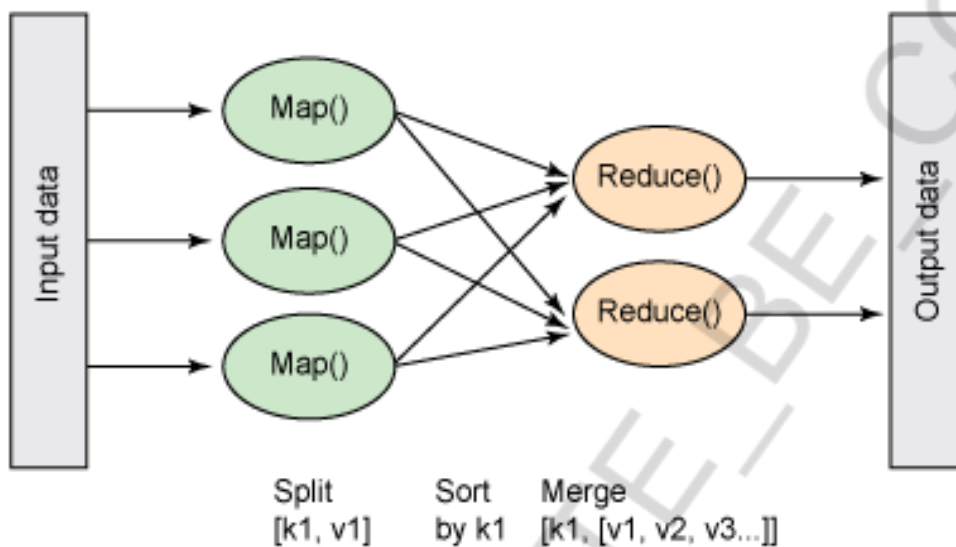
implementation known as Hadoop.

This project is used by **Yahoo, Facebook** and others.

To create a map function, the MapReduce architecture uses a functional programming style using key-value pair.

Reduce function merges all intermediate values using intermediate keys.

Hence programmers who do not have experience in parallel programming can simply use a large distributed processing environment without any problem.



Q.4 c) Identify the advantages and disadvantages of Direct Attached Storage?

Answer:

- **Advantage of DAS**
- DAS can provide users with **better performance** than **networked storage** because the server **does not have to traverse a network to read and write data**, which is why many organizations turn to DAS for applications that require high performance.
- DAS is also **less complex than network-based storage systems**, making it **easier to implement and maintain**, and **it is cheaper**.
- **Disadvantage of DAS**
- It has limited **scalability**
- lacks the type of centralized management
- **backup** capabilities available to other storage platforms.
- it can't be easily shared data

.....**End of Model Answer Sheet**.....